Binomial Theorem

Mind map

Basics and theorem

- $(x+y)^n = \sum {}^nC_r x^{n-r} y^r$
- $n+1$ terms, $\sum {}^nC_r = 2^n$
- Pascal's triangle

General term

- $T_{r+1} = {}^nC_r x^{n-r} y^r$
- Power of x = required, solve r
- Rational terms: r multiple of lcm

Middle term, products

- n even: $T_{\frac{n}{2}+1}$
- Greatest binomial coefficient
- Pair powers adding to required

Numerically greatest term

- $p = \frac{n+1}{1+\lvert a/b \rvert}$
- $T_{[p]+1}$; integer p: T_p, T_{p+1}
- T_k NGT: $k-1 < p < k$

Remainders

- Base = multiple $\pm$ small
- $a^n \pm b^n$ divisibility rules
- Last digits: mod 10^k

Integral and fractional part

- $R = I + f$, conjugate G
- $R + G \in \mathbb{Z}$, $RG = (a^2 - b)^n$
- I odd / even

Multinomial

- $\frac{n!}{r_1! r_2! r_3!}$, $\sum r_i = n$
- Dissimilar terms ${}^{n+k-1}C_{k-1}$
- Factorise the base

Series type 1 and 2

- Type 1: $\sum {}^nC_r x^{n-r} y^r = (x+y)^n$
- Type 2: $\sum {}^kC_r = {}^{m+1}C_{r+1}$
- Pascal's rule from first two terms

Series type 3

- Even / odd: $\frac{f(1) \pm f(-1)}{2}$
- $r\,{}^nC_r = n\,{}^{n-1}C_{r-1}$
- Gauss: reverse and add

Series type 4

- Factor r: differentiate
- Denominator $r+1$: integrate 0 to 1
- $\frac{{}^nC_r}{r+1} = \frac{{}^{n+1}C_{r+1}}{n+1}$

Series type 5

- Sum: $\sum {}^nC_r\,{}^mC_{k-r} = {}^{n+m}C_k$
- Difference: ${}^{n+m}C_{n+k}$
- Compare coefficient in $(1+x)^{n+m}$

Miscellaneous series

- GP of binomial terms
- ${}^nC_r = {}^nC_{n-r}$ pairing
- Reverse powers

Any index

- $\lvert x \rvert < 1$, infinite series
- $(1-x)^{-n}$: ${}^{n+r-1}C_r$
- Product: expand polynomial part

e^x and $\ln(1+x)$

- $e^x = \sum x^n/n!$
- $\ln(1+x) = x - \frac{x^2}{2} + \ldots$
- Split $n^2/n!$ into $n(n-1) + n$

Special tricks

- Ratio of consecutive coefficients
- $(x+a)^n \pm (x-a)^n$
- Approximation $1 + nx$